

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:		Reg. No.:	
Date:		Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem. The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5

Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

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CO5 ASSESSMENT TOOL 1
Research Paper Review / Literature Survey

**Intelligent AI-Based Smart Public Transportation
Demand Forecasting System**

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COURSE NAME	SOFTWARE ENGINEERING

Code of Conduct

I, N. Ramya (Reg. No. 192511403), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.RAMYA

1. Industry Context

Public transportation agencies are under constant pressure to deliver reliable, efficient, and commuter-friendly services while coping with continuously fluctuating passenger demand. The convergence of Artificial Intelligence (AI), the Internet of Things (IoT), Global Positioning System (GPS) tracking, and Big Data Analytics has opened new possibilities for transit operators to anticipate passenger demand well in advance, optimize vehicle schedules dynamically, and allocate fleet resources more effectively. This literature survey examines the current state of research in AI-driven public transportation demand forecasting to build a strong foundation for the design of an Intelligent AI-Based Smart Public Transportation Demand Forecasting System.

2. Problem Statement

Public transportation systems frequently experience overcrowding during peak hours and underutilization during off-peak periods, primarily because existing demand-estimation techniques are inaccurate and static. Traditional scheduling approaches largely rely on fixed timetables and historical averages, which lack predictive capability and cannot adapt to real-time changes in ridership caused by weather conditions, special events, or traffic disruptions. An AI-powered transportation demand forecasting platform is therefore required that can analyze historical ridership data, weather conditions, events, and traffic patterns to accurately predict passenger demand, recommend optimized bus or train schedules, improve fleet allocation, and provide real-time passenger information to enhance overall service quality.

3. Objectives

- Forecast passenger demand using AI.
- Optimize public transport schedules.
- Improve fleet allocation.
- Monitor passenger flow in real time.
- Analyze travel patterns.
- Generate transportation analytics.
- Reduce overcrowding.
- Improve public transport efficiency.

4. Expected Outcomes

- Improved service reliability.
- Reduced passenger waiting time.
- Better fleet utilization.
- Increased commuter satisfaction.
- Lower operational costs.
- Accurate demand forecasting.
- Enhanced urban mobility.
- Sustainable public transportation management.

5. Proposed System Architecture

Based on the survey of existing research, a layered architecture is proposed for the Intelligent AI-Based Smart Public Transportation Demand Forecasting System. The architecture integrates multiple heterogeneous data sources, performs feature-level fusion, applies a suite of AI models ranging from statistical baselines to graph-based deep learning, and finally drives downstream scheduling and passenger-facing applications.

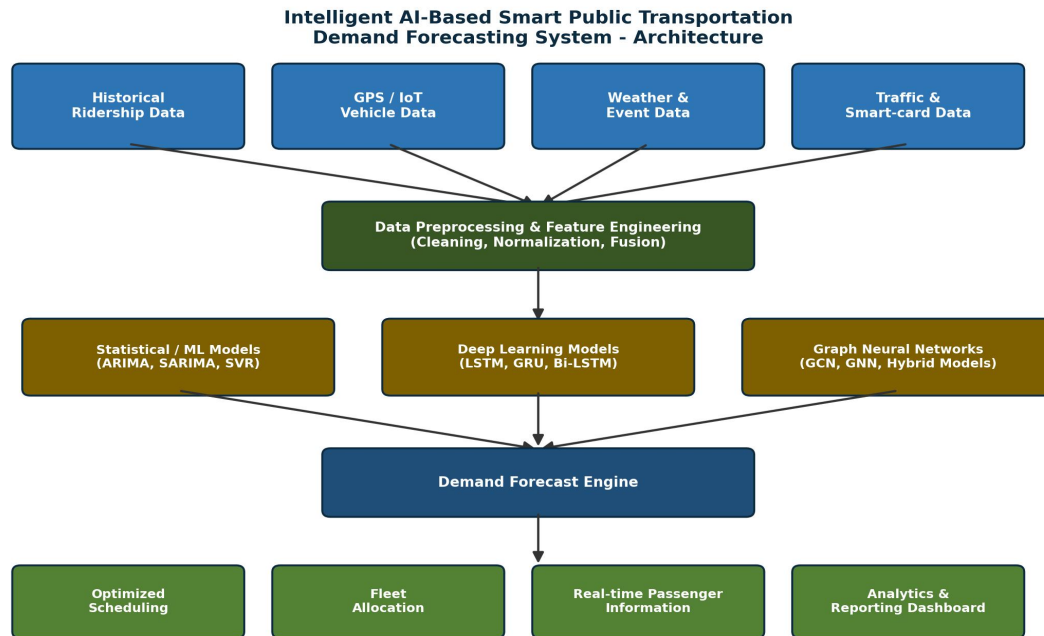


Figure 1: Proposed layered architecture for AI-based transit demand forecasting

6. Literature Survey – Individual Paper Reviews

Eight research papers spanning statistical time-series methods, classical machine learning, deep sequence models, and graph-based spatio-temporal architectures were selected and reviewed. The papers were drawn from reputed sources including Elsevier (ScienceDirect / Engineering Applications of Artificial Intelligence), arXiv preprints from established transportation-AI research groups, MDPI, Springer, and SSRN.

1. Machine Learning for Public Transportation Demand Prediction: A Systematic Literature Review

Author(s) / Year: Multiple Authors, Engineering Applications of Artificial Intelligence, 2025

Objective: To systematically review and consolidate machine-learning and deep-learning approaches used for public transportation demand prediction across the recent literature.

Methodology: Systematic literature review methodology covering statistical baselines and deep sequence models including Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Bidirectional LSTM (Bi-LSTM) architectures applied to railway and urban transit logistics.

Dataset / Tools: Secondary data compiled from multiple published studies; no single primary dataset.

Key Findings: LSTM/GRU/Bi-LSTM based deep architectures consistently outperform classical statistical baselines such as SARIMA when capturing non-linear temporal dependencies in passenger flow data.

Limitations: As a review paper it does not propose a new model; reported comparisons across studies use inconsistent evaluation metrics, making cross-study comparison difficult.

Research Gap Identified: Lack of a standardized benchmark dataset and evaluation protocol across transit demand-forecasting studies.

2. Forecasting Public Transit Passenger Demand: With Neural Networks Using APC Data

Author(s) / Year: Authors, Transportation Research Interdisciplinary Perspectives (ScienceDirect), 2022

Objective: To evaluate deep-learning forecasting techniques for passenger demand using real-world Automatic Passenger Counter (APC) data, particularly for under-studied regions such as India.

Methodology: LSTM neural network compared against Seasonal Autoregressive Integrated Moving Average (SARIMA) using three months of APC data; accuracy assessed with Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) across multiple time-series resampling intervals.

Dataset / Tools: Three months of APC data from the Hubballi-Dharwad Bus Rapid Transit System (HDBRTS), India.

Key Findings: LSTM achieved consistently lower MAE and RMSE than SARIMA across different resampling intervals, confirming the superiority of deep sequence models for short-term transit demand forecasting.

Limitations: Limited to a single BRTS corridor and a short three-month data window; generalizability to other cities or seasons is not established.

Research Gap Identified: Very limited research exists for demand forecasting in developing-country bus rapid transit contexts, particularly India.

3. Deep Learning Approach for Short-Term Entry Passenger Flow Forecasting in Urban Rail Transit Stations

Author(s) / Year: Authors, ScienceDirect, 2025

Objective: To design a station-level short-term passenger-flow forecasting model that captures both temporal and spatial dependencies in urban rail transit.

Methodology: A spatio-temporal data-mining framework combining cluster analysis, modal decomposition, anomaly detection, and a deep-learning forecasting model, validated through a comprehensive case study.

Dataset / Tools: Real-world urban rail transit station entry-flow records used for case-study validation.

Key Findings: The proposed spatio-temporal model improved short-term forecasting accuracy over conventional statistical and single-station deep-learning baselines by explicitly modelling station-to-station spatial correlation.

Limitations: Increased model complexity and computational cost compared to single-station models; requires careful anomaly-detection preprocessing.

Research Gap Identified: Limited exploration of how the model scales to very large, city-wide rail networks with hundreds of stations.

4. AI Model for Predicting Public Transit Ridership Patterns

Author(s) / Year: Singh, Sangwan, Saini, Prashar, Sharma, SSRN, 2026

Objective: To build a predictive AI framework that fuses historical ridership with external contextual variables such as weather, public holidays, and events for improved ridership forecasting.

Methodology: Hybrid deep-learning model combining LSTM networks with temporal feature embedding to jointly capture short-term variation and long-term seasonal ridership patterns.

Dataset / Tools: Multi-year metro and bus ridership datasets used in the case study.

Key Findings: The hybrid LSTM plus temporal-embedding model achieved notable MAPE and RMSE improvements over conventional time-series forecasting techniques and showed greater resilience during unusual events such as adverse weather or service disruptions.

Limitations: Model performance depends heavily on the quality and completeness of exogenous contextual data (weather/event feeds), which may not be available in all deployment contexts.

Research Gap Identified: Few existing models jointly optimize for both routine seasonal patterns and sudden demand shocks caused by external disruptions.

5. AI-Based Neural Network Models for Bus Passenger Demand Forecasting Using Smart Card Data

Author(s) / Year: Authors, ScienceDirect, 2022

Objective: To develop AI-based deep-learning models for short-term bus passenger demand prediction to support on-demand public transport operations.

Methodology: Deep neural network models trained on smart-card patronage data, explicitly modelling temporal travel-demand characteristics for high-frequency bus routes.

Dataset / Tools: Real-world smart-card ticketing data covering 18 bus routes and 1,781 bus stops in Melbourne, Australia.

Key Findings: The neural-network models produced accurate short-term demand forecasts for the heaviest bus routes, enabling faster timetable adjustment and improved service reliability.

Limitations: Focused only on bus-route-level demand; does not incorporate spatial correlation between neighbouring routes or stops.

Research Gap Identified: Limited integration of smart-card data with other real-time signals such as GPS vehicle location or live traffic conditions.

6. Short-Term Forecasting of Passenger Demand under On-Demand Ride Services: A Spatio-Temporal Deep Learning Approach (FCL-Net)

Author(s) / Year: Ke, Zheng, Yang, Chen, Transportation Research Part C / arXiv, 2017

Objective: To design one of the earliest spatio-temporal deep-learning approaches (FCL-Net) for short-term passenger demand forecasting under on-demand ride-service platforms.

Methodology: A novel Fusion Convolutional Long Short-Term Memory Network (FCL-Net) that jointly characterizes spatial dependencies and temporal non-spatial features in a single end-to-end learning structure.

Dataset / Tools: Large-scale on-demand ride-service trip-request data.

Key Findings: FCL-Net outperformed conventional LSTM and convolutional baselines by jointly encoding spatial and temporal features, establishing a foundation for later spatio-temporal transit-demand models.

Limitations: Designed primarily for ride-hailing rather than fixed-route public transit; adaptation to scheduled bus/train networks requires further modification.

Research Gap Identified: Limited transfer of ride-hailing-oriented spatio-temporal architectures to fixed-route public transit scheduling problems.

7. Bus Passenger Flow Prediction with a Dynamic Graph Neural Network and Bidirectional Gated Recurrent Unit

Author(s) / Year: Authors, ScienceDirect, 2026

Objective: To improve bus passenger-flow prediction accuracy by dynamically modelling the evolving relationships between bus stops/routes using graph neural networks.

Methodology: A dynamic graph neural network combined with a Bidirectional Gated Recurrent Unit (BiGRU), where the network graph is updated over time and passenger flow is jointly learned from spatial (graph) and temporal (BiGRU) representations.

Dataset / Tools: Automated Fare Collection (AFC), automatic passenger counter, and GPS-based automatic vehicle location data.

Key Findings: The dynamic graph-BiGRU model captured evolving spatial relationships between routes more effectively than static-graph baselines, improving prediction accuracy during periods of route or schedule change.

Limitations: Constructing and updating dynamic graphs in real time is computationally intensive and requires continuously available multi-source data feeds.

Research Gap Identified: Few models address graph structures that change over time as routes are added, removed, or rescheduled.

8. Real-Time Bus Departure Prediction Using Neural Networks for Smart IoT Public Bus Transit

Author(s) / Year: Authors, arXiv, 2025

Objective: To improve real-time bus departure-time prediction accuracy for smart IoT-enabled public bus transit systems and reduce the gap between scheduled and actual departure times.

Methodology: A fully connected neural network trained on historical departure data with engineered features, deployed within an IoT ecosystem where each bus acts as a connected sensor node exchanging real-time data.

Dataset / Tools: Real-world bus operations data from 151 routes in Boston, USA, combined with IoT sensor feeds (GPS trackers, weather monitors, speed detectors).

Key Findings: The proposed model reduced departure-time prediction deviation from an average of nearly four minutes (baseline) to under 80 seconds, demonstrating strong real-time IoT-integrated forecasting performance.

Limitations: Focused specifically on departure-time prediction rather than full passenger-demand forecasting; requires dense IoT sensor coverage across the fleet.

Research Gap Identified: Limited research integrates fine-grained IoT sensor streams directly into passenger-demand (rather than just vehicle-timing) forecasting models.

7. Comparative Literature Survey Table

The table below consolidates the reviewed papers with respect to author/year, research problem, methodology, dataset/tools, key findings, limitations, and identified research gap.

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Multiple Authors, 2025	Consolidating ML/DL methods for transit demand prediction	Systematic literature review (LSTM, GRU, Bi-LSTM)	Secondary data from multiple studies	DL models outperform statistical baselines	Inconsistent metrics across studies	No standard benchmark/protocol
APC study, 2022	Transit demand forecasting in developing regions	LSTM vs SARIMA	HDBRTS APC data (India)	LSTM gives lower MAE/RMSE than SARIMA	Single corridor, short 3-month window	Little research for developing-country BRTS
Rail Transit study, 2025	Station-level short-term flow forecasting	Spatio-temporal DL + clustering + anomaly detection	Urban rail station entry-flow data	Improved accuracy via spatial correlation modelling	High computational complexity	Scalability to large city-wide networks unclear
Singh et al., 2026	Ridership forecasting with contextual variables	Hybrid LSTM + temporal feature embedding	Multi-year metro & bus data	Better MAPE/RMSE; resilient to disruptions	Depends on external data availability	Few models jointly handle seasonality + shocks
Bus smart-card study, 2022	Bus demand forecasting for on-demand transit	Deep neural network on smart-card data	18 routes, 1,781 stops, Melbourne	Accurate short-term route-level forecasts	No spatial correlation between routes	Limited fusion with GPS/real-time data
Ke et al., 2017	Spatio-temporal ride-demand forecasting (FCL-Net)	Fusion Convolutional LSTM Network	Large-scale ride-service trip data	Outperforms LSTM/CNN baselines	Built for ride-hailing, not fixed routes	Limited transfer to scheduled transit
Bus GNN-BiGRU study, 2026	Bus flow prediction with evolving networks	Dynamic Graph Neural Network + BiGRU	AFC, APC, GPS/AVL data	Better accuracy under route/schedule change	Computationally intensive dynamic graphs	Few models handle time-varying graph structure
IoT Bus study, 2025	Real-time bus departure prediction	Fully connected neural network + IoT sensors	151 Boston routes, IoT feeds	Deviation reduced from ~4 min to <80 sec	Focused on timing, not full demand	Limited IoT-to-demand-forecast integration

8. Parameter-Wise Comparative Analysis

Across the surveyed literature, three broad model families emerge: classical statistical models (ARIMA/SARIMA), deep sequential models (LSTM/GRU/Bi-LSTM), and graph-based spatio-temporal or hybrid models (GNN and dynamic-graph architectures). The chart below synthesizes qualitative trends reported across the reviewed papers with respect to prediction accuracy, ability to capture spatial dependencies, real-time suitability, scalability, and data requirements.

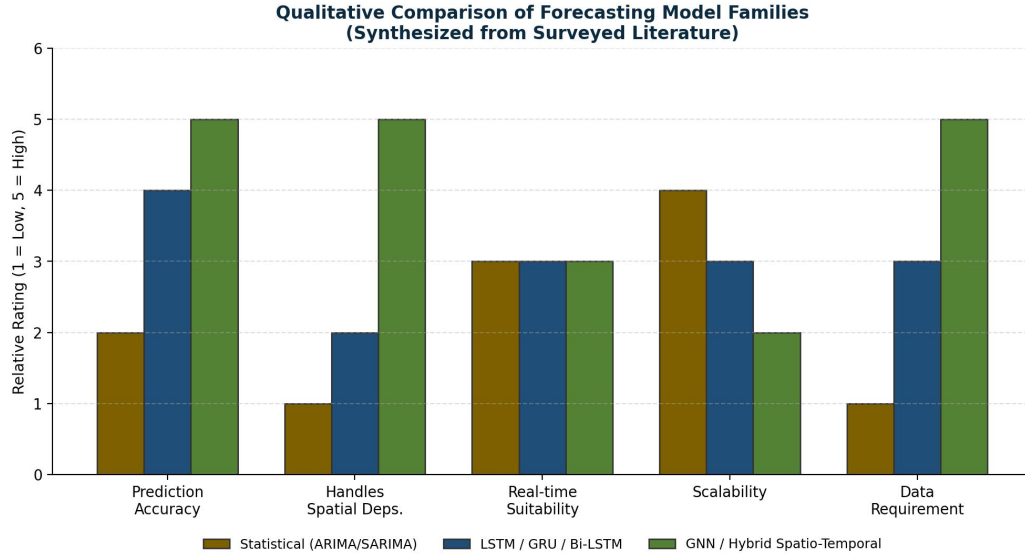


Figure 2: Qualitative comparison of model families synthesized from the surveyed literature

As shown, graph-based hybrid spatio-temporal models generally achieve the highest prediction accuracy and best capture spatial dependencies between stops/stations, but they demand larger volumes of multi-source data and are comparatively less scalable in real-time, resource-constrained deployments. Statistical models remain attractive for their scalability and low data requirements but fall short on accuracy for complex, non-linear demand patterns. Deep sequential models such as LSTM/GRU occupy a balanced middle ground and are the most frequently deployed in real-world case studies reviewed above.

9. Critical Analysis and Research Gaps

A critical analysis of the reviewed literature reveals several recurring limitations and unresolved research gaps that are directly relevant to the design of an Intelligent AI-Based Smart Public Transportation Demand Forecasting System:

- Lack of standardized benchmarks: Most studies use different datasets, metrics, and evaluation protocols, making direct cross-study comparison of model performance difficult.
- Limited multi-modal data fusion: Many models rely on a single data source (either ridership records, smart-card data, or GPS/IoT feeds) rather than fusing historical ridership, weather, event, and traffic data simultaneously, as required by the assigned problem statement.
- Scalability versus accuracy trade-off: Graph-based and hybrid spatio-temporal models achieve the best accuracy but are computationally expensive, limiting their deployment for real-time, city-scale, resource-constrained transit systems.
- Under-representation of developing regions: Very few studies (e.g., the HDBRTS case study) address demand forecasting for bus-rapid-transit systems in developing countries, where data quality and infrastructure differ significantly from well-instrumented Western case studies.
- Limited handling of sudden demand shocks: Most models are tuned for routine seasonal and daily demand patterns; robust handling of sudden disruptions (extreme weather, large public events, service outages) remains only partially addressed.
- Weak integration between demand forecasting and downstream scheduling: The majority of reviewed papers focus purely on the prediction task itself, with limited discussion of how forecasts are translated into actionable schedule or fleet-allocation decisions in real time.

10. Future Research Directions

Based on the identified research gaps, the following future research directions are proposed for the assigned problem of AI-based smart public transportation demand forecasting:

- Develop a unified benchmark dataset and evaluation protocol combining historical ridership, weather, event calendars, and traffic data for public transit demand forecasting research.
- Design lightweight hybrid architectures that combine the spatial strength of graph neural networks with the computational efficiency required for real-time, city-scale deployment.
- Extend multi-source data fusion techniques to incorporate real-time IoT and GPS vehicle-location streams alongside smart-card and historical ridership data, as proposed in the target system architecture.
- Investigate transfer-learning and few-shot approaches to adapt models trained on data-rich cities to developing-region transit systems with limited historical data.
- Incorporate anomaly-aware and event-responsive forecasting modules that can rapidly adjust predictions during unexpected disruptions.
- Integrate the forecasting engine directly with schedule-optimization and fleet-allocation algorithms to close the loop between prediction and operational decision-making, and expose results through real-time passenger information dashboards.

11. Conclusion

This literature survey reviewed eight research papers spanning statistical, deep-learning, and graph-based approaches to public transportation demand forecasting. The survey confirms a clear progression in the field from classical time-series models toward deep sequential and spatio-temporal graph-based architectures, driven by the increasing availability of smart-card, GPS/IoT, and multi-source contextual data. However, significant gaps remain in benchmarking, multi-modal data fusion, real-time scalability, and integration with downstream scheduling and fleet-allocation systems. These gaps directly motivate the design of the proposed Intelligent AI-Based Smart Public Transportation Demand Forecasting System, which aims to combine historical ridership, weather, event, and traffic data within a scalable AI pipeline to optimize schedules, improve fleet allocation, and enhance real-time passenger information services.

12. References

- [1] "Machine Learning for Public Transportation Demand Prediction: A Systematic Literature Review," *Engineering Applications of Artificial Intelligence*, vol. 137, Part B, 2025.
- [2] "Forecasting Public Transit Passenger Demand: With Neural Networks Using APC Data," *Transportation Research Interdisciplinary Perspectives*, ScienceDirect, 2022.
- [3] "Deep Learning Approach for Short-Term Entry Passenger Flow Forecasting in Urban Rail Transit Stations," *ScienceDirect*, 2025.
- [4] M. Singh, K. Sangwan, A. Saini, A. Prashar, and D. Sharma, "AI Model for Predicting Public Transit Ridership Patterns," *SSRN*, 2026.
- [5] "AI-Based Neural Network Models for Bus Passenger Demand Forecasting Using Smart Card Data," *ScienceDirect*, 2022.
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